Aman Yadav

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Summary

Mathematics & Computing undergraduate focused on distributed systems, high-throughput API gateways, and asynchronous backend architectures. Experienced in building sub-millisecond edge caching, database-backed worker fleets, and resilient REST APIs using Python, FastAPI, PostgreSQL, Redis, Celery, and Docker.

Education

Central University of Karnataka — Kalaburagi, Karnataka, India
Bachelor of Technology in Mathematics & Computing — *2023 – 2027*

- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Distributed Computing, Object-Oriented Programming, Linear Algebra, Probability & Statistics, Numerical Optimization.

Experience

Maruti Suzuki India Limited — Gurugram, India
Engineering Intern (Applied ML & Telematics) – EN-DAX Dept — *May 2026 – Jul 2026*

- Developed a 6-stage ingestion pipeline normalizing **1M+ raw CAN-bus records** into a **40-feature automotive taxonomy** with **automated trip segmentation**.
- Derived vehicle kinematics using vectorized **Haversine trajectory metrics** and speed differentials to classify harsh driving events, applying dual-stage **IQR and Z-score outlier filtering**.
- Delivered an interactive Streamlit analytics console with **automated JSON audit trails** and validation workflows, enabling vehicle dynamics engineers to calibrate sensor threshold distributions.

Projects

PingGuard – Distributed HTTP Uptime Monitoring Engine — GitHub Repo
FastAPI, PostgreSQL, Redis, Celery, SQLAlchemy, Docker

- Decoupled the API control plane from outbound probing I/O via **distributed worker fleets** with **exponential backoff recovery** to mitigate downstream network timeouts.
- Engineered database-driven task dispatching using **row-level locking** (`SELECT ... FOR UPDATE SKIP LOCKED`) with **transactional recovery** to eliminate race conditions and task loss.
- Hardened network probing against **SSRF and TOCTOU DNS rebinding** via **asynchronous pre-flight DNS validation** against private CIDR ranges with **direct IP pinning**.

GateForge – High-Throughput API Gateway & Rate Limiting Engine — GitHub Repo
FastAPI, PostgreSQL, Redis, Lua, Docker, Prometheus

- Architected a two-tier edge authentication gateway with **zero plaintext secret storage** via **SHA-256 cryptographic hashing**, achieving $<$**1ms verification latency** using edge RAM caching with **anti-stampede random jitter**.
- Engineered an **atomic sliding-window rate limiter** via Redis Lua scripts (`ZREMRANGEBYSCORE/ZADD`) to eliminate window-edge bursts; enforced **RFC 6585 rate-limit headers** and **CIDR/IP allowlisting**.
- Built an **asynchronous non-blocking telemetry engine** buffering request events in bounded in-memory queues and flushing batched DB writes; exposed **Prometheus p50/p95/p99 latency histograms** and **circuit breaker trip states**.

EduPulse – Academic Forecasting & Early-Warning Platform — GitHub Repo
Django, PostgreSQL, Redis, Celery, Scikit-Learn, Docker

- Architected a modular backend implementing **fine-grained RBAC**, **JWT authentication**, and **strict ACID transactional guarantees** across relational schemas.
- Engineered an **isolated dual-instance Redis topology** separating task queues from caching, leveraging **generational cache invalidation** to sustain $<$**40ms response times**.
- Built batch inference pipelines with **GroupedKFold cross-validation** and **demographic feature quarantine** to prevent bias leakage; maintained system stability with **220+ automated tests**.

Technical Skills

Languages: Python, SQL (PostgreSQL), C++, Lua, Bash

Backend & Distributed Systems: FastAPI, Django, Celery, Redis, RESTful APIs, AsyncIO, Microservices, Rate Limiting, RBAC/JWT

Databases & Storage: PostgreSQL, Redis (Caching & Message Queues), SQLAlchemy, Alembic, Relational Schema Design

DevOps & Observability: Docker, Docker Compose, Linux, Prometheus, Git, GitHub Actions, Postman, pytest

Data & ML: scikit-learn, Pandas, NumPy, Streamlit